Experiment No: 9

Aim : A). Create a Lexer to take input from text file and count no of characters, no. of lines & no. of words.

Program :

```
%{
#include<stdio.h>
 int lines=0, words=0,s_letters=0,c_letters=0, num=0, spl_char=0,total=0;
%}
%%
\n { lines++; words++;}
[\t ' '] words++;
[A-Z] c_letters++;
[a-z] s_letters++;
[0-9] num++;
. spl_char++;
%%
main(void)
{
yyin= fopen("practical9.txt","r");
yylex();
total=s_letters+c_letters+num+spl_char;
printf(" This File contains ...");
printf("\n\t%d lines", lines);
printf("\n\t%d words",words);
printf("\n\t%d small letters", s_letters);
printf("\n\t%d capital letters",c_letters);
printf("\n\t%d digits", num);
printf("\n\t%d special characters",spl_char);
 printf("\n\tIn total %d characters.\n",total);
}
int yywrap()
{
return(1);
}
```

Practical9.txt FILE :

Hii
I'm Dijay From Cse

Output :

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19042.804]
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D:\6th Sem\Compiler Design\LexAndYacc>flex practical9.l

D:\6th Sem\Compiler Design\LexAndYacc>gcc lex.yy.c

D:\6th Sem\Compiler Design\LexAndYacc>a.exe
 This File contains ...
        1 lines
        7 words
        12 small letters
        5 capital letters
        0 digits
        0 special characters
        In total 17 characters.

D:\6th Sem\Compiler Design\LexAndYacc>
```

B). Write a Lex program to count number of vowels and consonants in a given input string.

Program :

```
%{
 int vow_count=0;
int const_count =0;
%}
%%
[aeiouAEIOU] {vow_count++;}
[a-zA-Z] {const_count++;}
%%
main()
{
yyin= fopen("practical8.txt","r");
yylex();
 printf("The number of vowels are: %d\n",vow_count);
printf("The number of consonants are: %d\n",const_count);
return 0;
}
yywrap()
{
return 1;
 }
```

practical8.txt FILE

Hii

Experiment No: 10

Aim : A). Write a Lex program to print out all numbers from the given file.

Program :

```
%{
#include<stdio.h>
int num=0;
%}
%%
[0-9] num++; ECHO;
%%
main(void)
{
yyin= fopen("practical10.txt","r");
yylex();
printf("\n\t%d  digits",  num);
}
int yywrap()
{
return(1);
}
```

Practical10.txt FILE

23332
23
Sbfjehbcc

Output :

```
C:\Windows\System32\cmd.exe

D:\6th Sem\Compiler Design\LexAndYacc>flex practical10.l

D:\6th Sem\Compiler Design\LexAndYacc>gcc lex.yy.c

D:\6th Sem\Compiler Design\LexAndYacc>a.exe
23332
23
sbfjehbcc
        7  digits
D:\6th Sem\Compiler Design\LexAndYacc>
```

B). Write a Lex program to printout all HTML tags in file.

Program :

```
%{
#include<stdio.h>
%}

%%
\<[^>]*\> printf("%s\n",yytext);
.|\n;
%%

int yywrap()
{
return 1;
}

int main()
{
yyin=fopen("practical10b.txt","r");
yylex();
return 0;
}
```

Practical10b.txt FILE

```
<html>
<body>
<h1> Hello</h1>
<br>
</body>
</html>
```

Output :

```
C:\Windows\System32\cmd.exe

D:\6th Sem\Compiler Design\LexAndYacc>flex practical10b.l

D:\6th Sem\Compiler Design\LexAndYacc>gcc lex.yy.c

D:\6th Sem\Compiler Design\LexAndYacc>a.exe
<html>

<body>

<h1>
</h1>

<br>

</body>

</html>
```

C). Write a Lex program which adds line numbers to the given file and display the same onto the standard output.

Program :

```
%{
        #include<stdio.h>
        int line_number = 1;
%}
line .*\n
%%
{line} { printf("%d %s", line_number++, yytext); }
%%
int yywrap(){}
int main()
{
        extern FILE *yyin;
        yyin=fopen("add.txt","r");
        if(yyin==NULL){
                printf("File Not Found");
        }
        yylex();
        return 0;
}
```

Add.txt FILE

```
#include<stdio.h>
int main()
{
int a=10,b=20;
int c=a+b;
printf("Addition of a and b is:%d",c);
return 0;
}
```

Output :

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19042.804]
(c) 2020 Microsoft Corporation. All rights reserved.

D:\6th Sem\Compiler Design\All Programs\2>flex AddLineNumber.l

D:\6th Sem\Compiler Design\All Programs\2>gcc lex.yy.c

D:\6th Sem\Compiler Design\All Programs\2>a.exe
1 #include<stdio.h>
2 int main()
3 {
4 int a=10,b=20;
5 int c=a+b;
6 printf("Addition of a and b is:%d",c);
7 return 0;
8 }

D:\6th Sem\Compiler Design\All Programs\2>
```